

Julien Colin

ELLIS PhD Student

PhD Student

ELLIS Alicante

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Education

- 2022–present **PhD, Computer Science**, *ELLIS Alicante Foundation, University of Alicante, Spain.*
Human-centered Interpretability, under the supervision of Nuria Oliver, Thomas Serre and Miguel Angel Lozano
- 2020–2021 : **Master 2 Cognitive Sciences, Natural and Artificial Cognition**, *Grenoble Institute of Technology (INP), France.*
- 2019–2020 : **Master 1 Theoretical and Applied Physics, Biophysics**, *Paul Sabatier University, France.*
- 2014–2019 : **Bachelor of Science: Physics and Chemistry**, *University of Lorraine, France.*

Research Experience

[Brown University, USA](#)

- April, 2025 – **Visiting Research Fellow.**
Oct 2025 Human alignment and interpretability in visual representations.

[Brown University, USA](#)

- April, 2022 – **Research Associate.**
Sept 2022 Development of methods and metrics for Explainable Artificial Intelligence.

[ANITI, France](#)

- Sept, 2021 – **Research Associate.**
Feb 2022 Development of methods and metrics for Explainable Artificial Intelligence.

[ANITI, France](#)

- Feb, 2021 – **Research Intern.**
July 2021 Development of an evaluation framework for Explainable Artificial Intelligence methods.

[Animal Cognition Research Center \(CRCA\), France](#)

- May, 2020 – **Research Intern.**
June 2020 Modeling of the movement of bumblebees through Lattice Boltzmann methods.

[French National Centre for Scientific Research \(CNRS\), France](#)

- April, 2019 – **Research Intern.**
May 2019 Coarse-grained simulation of DNA-proteins repair complex in the cellular environment.

Publications

[Work in Progress](#)

- 2026 **Colin, J.**, Goetschalckx, L., Fel, T., Boutin, V., Serre, T., Oliver, N., Choosing the right basis for interpretability: Psychophysical comparison between neuron-based and dictionary-based representations., *Under review at TMLR.*

- 2026 **Colin, J.**, Goetschalckx, L., Oliver, N., Serre, T., Capability $\neq$ Interpretability: Human Interpretability of Vision Foundation Models., *Under review at NeurIPS*.

In Conference Proceedings

- 2026 **Colin, J.**, Oliver, N., Serre, T., Can human perceptual similarity alignment improve interpretability in visual representations?, Conference on Cognitive Computational Neuroscience. (CCN)
- 2026 Colin, J., Oliver, N., and Serre, T. , Does human-alignment benefit interpretability?, Workshop on Representational Alignment (Re⁴-Align)(ICLR).
- 2023 Fel, T., Boissin, T., Boutin, V., Picard, A., Novello, P., **Colin, J.**, Linsley, D., Rousseau, T., Cadène, R., Gardes, L., and Serre, T., Unlocking Feature Visualization for Deeper Networks with MAGnitude Constrained Optimization., Neural Information Processing Systems (**NeurIPS**).
- 2023 Boutin, V., Fel, T., Singhal, L., Mukherji, R., Nagaraj, A., **Colin, J.**, and Serre, T., Diffusion models as artists: Are we closing the gap between humans and machines?, International Conference on Machine Learning (**ICML**).
- 2023 Fel, T., Picard, A., Bethune, L., Boissin, T., Vigouroux, D., **Colin, J.**, Cadène, R., and Serre, T. , CRAFT: Concept Recursive Activation FacTORIZATION for explainability, IEEE Conference on Computer Vision and Pattern Recognition (**CVPR**).
- 2022 **Colin, J.**, Fel, T., Cadène, R., and Serre, T., What I cannot predict, I do not understand: A human-centered evaluation framework for explainability methods., Neural Information Processing Systems (**NeurIPS**).
- 2022 Zerroug, A., Vaishnav, M., **Colin, J.**, Musslick, S., and Serre, T. , A benchmark for compositional visual reasoning., Neural Information Processing Systems. Dataset and Benchmarks (**NeurIPS**).
- 2022 Fel, T., Hervier, L., Vigouroux, D., Poche, A., Plakoo, J., Cadène, R., Chalvidal, M., **Colin, J.**, Boissin, T., Bethune, L., Picard, A., Nicodeme, C., Gardes, L., Flandin, G., and Serre, T., Xplique: A deep learning explainability toolbox., IEEE Conference on Computer Vision and Pattern Recognition. Workshop on XAI4CV: Explainable Artificial Intelligence for Computer Vision (**CVPR**).

Journal Article

- 2024 Riccio, P., **Colin, J.**, Ogolla, S., Oliver, N., Mirror, Mirror on the Wall, Who Is the Whitest of All? Racial Biases in Social Media Beauty Filters, In ***Social Media + Society***.

Scientific Service Community

- 2022-25 **Co-chair**, ELLIS Reading Group on Human-Centric Machine Learning.
- 2023-24-25-
26 **Reviewer**, NeurIPS.
- 2024-25-26 **Reviewer**, ICLR: workshop Re-Align.
- 2026 **Reviewer**, CCN.
- 2025 **Reviewer**, AAIL: AI Alignement track.
- 2024 **Reviewer**, NeurIPS: workshop Behavioral ML.
- 2024 **Reviewer**, ECCV: workshop XAI4CV.
- 2023 **Reviewer**, CVPR: workshop XAI4CV.
- 2024 **Reviewer**, ICLR.
- 2024 **Reviewer**, ICML.
- 2023-24 **Reviewer**, ICCV/ECCV.

2023 **Program committee and Reviewer**, ECAI.

2022 **Reviewer**, CVPR.

Computer skills

Programming Languages Python (PyTorch), Javascript